GUJARAT TECHNOLOGICAL UNIVERSITY
K. D. POLYTECHNIC, PATAN
DEPARTMENT OF COMPUTER ENGINEERING
Academic Year: 2026-27 | Semester: V (Diploma Engineering)

MINOR PROJECT LOGBOOK

Team Details

Sr. No.	Enrollment Number	Student Name	Class	Batch
1	246310307123	**PATEL SHUBH MAHESHKUMAR (Team Lead)**	**B**	**B1**
2	246310307028	**DAVE KRUSHANK JIGNESHBHAI**	**A**	**A2**
3	246310307090	**PATEL FIYANS BABUBHAI**	**A**	**A4**
4	246310307147	**PRAJAPATI VEER HEMENDRAKUMAR**	**B**	**B2**

Note: Fill only the required number of rows as per actual team size.

Project Details

Project Title:

DineFlow AI: Next-Generation Restaurant Cloud POS, Contactless QR Ordering, Real-Time KDS & Smart ERP System

Technologies: Laravel 11, MySQL, Python AI Engine, Electron Desktop App, Mobile Web PWA, Real-time KDS

Project Guide Details & Signature

Name of Project Guide: **Prof. Yagnesh Patel** Lecturer, Department of Computer Engineering K. D. Polytechnic, Patan	**Signature & Approval:** _____ Date: _____ / _____ / 2026

WEEKLY LOGBOOK RECORD (WEEK 1 TO WEEK 8)

Week No.	Date Duration	Work Planned	Work Completed	Remark	Guide Sign
1	06/07 - 11/07	Problem identification, domain research, and restaurant operation workflow study.	Surveyed 3 local dining restaurants in Patan; identified order delays and paper ticket loss.	Approved	
2	13/07 - 18/07	Literature survey of existing POS systems and formulation of project objectives.	Benchmarked Petpooja and Toast POS; finalized scope of cloud multi-tenant architecture.	Approved	
3	20/07 - 25/07	Software Requirement Specification (SRS) documentation and entity identification.	Completed SRS with IEEE format; defined functional requirements and customer roles.	Satisfactory	
4	27/07 - 01/08	System architecture design, DFD Level 0/1/2, and technical stack finalization.	Created system flowcharts, DFDs, and state machine diagrams; selected Laravel 11 & MySQL.	Approved	
5	03/08 - 08/08	Relational database design, 3NF normalization, and ER diagram drafting.	Designed 14 normalized database tables with foreign key constraints and audit timestamps.	Good	
6	10/08 - 15/08	Database migrations, seeders setup, and REST API routing scaffold.	Executed Laravel migrations, created seeder datasets for 50 dishes, and configured REST API.	Satisfactory	
7	17/08 - 22/08	Touchscreen billing terminal UI development and category menu grid.	Built responsive cashier POS grid, item search bar, and dynamic bill calculations.	Approved	
8	24/08 - 29/08	Mid-Semester project demonstration, code review, and guide evaluation.	Presented working cashier POS terminal to guide; incorporated split-bill suggestions.	Very Good	

WEEKLY LOGBOOK RECORD (WEEK 9 TO WEEK 15)

Week No.	Date Duration	Work Planned	Work Completed	Remark	Guide Sign
9	31/08 - 05/09	Customer contactless QR menu interface and table session isolation.	Built mobile customer menu with veg/non-veg filters, dish cards, and local cart storage.	Approved	
10	07/09 - 12/09	Real-time Kitchen Display System (KDS) & audio alert notification engine.	Implemented live chef order board with auto-refresh and Gujarati/Hindi audio chime.	Satisfactory	
11	14/09 - 19/09	Offline sync engine & Electron Windows desktop app encapsulation.	Packaged Electron wrapper with local SQLite cache and backoff background sync engine.	Good	
12	21/09 - 26/09	Recipe inventory management, auto raw-material stock depletion & alerts.	Added ingredient BOM link to menu items; tested automatic stock deduction on orders.	Approved	
13	28/09 - 03/10	SuperAdmin SaaS analytics, restaurant onboarding & subscription engine.	Built multi-restaurant admin suite, daily revenue charts, and billing analytics.	Satisfactory	
14	05/10 - 10/10	Thermal receipt printing, A4 tax invoice generation & WhatsApp dispatch.	Engineered 80mm ESC/POS thermal printer layout, PDF A4 invoice, and WhatsApp modal.	Approved	
15	12/10 - 17/10	Cloud VPS production deployment, performance hardening & final report.	Deployed to AWS EC2 with Nginx, SSL, and domain; completed final GTU project report.	Excellent	

End-of-Semester Project Completion Verification:

Project Guide Completion Sign-off:

Prof. Yagnesh Patel

Lecturer, Department of Computer Engineering, K. D. Polytechnic, Patan

Status: 15-Week Project Logbook Verified & Completed for Semester End Submission.

Guide Signature:

_____

Date: _____ / _____ / 2026

ANNEXURE: OFFICIAL GITHUB REPOSITORY & CODE CONTRIBUTION SUMMARY

DineFlow AI: Minor Project (DI05000341) | Git Verification Record

1. GitHub Repository Overview & Verification Details

Public GitHub Repository	**https://github.com/Shubh-Patel-07/DineFlow-AI**
Repository Owner / Account	Shubh-Patel-07 (Patel-Shubh-07)
Total Verified Commits	**617 Commits** on main branch (1,532+ across agile feature branches)
Active Feature Branches	main, feature-teamlead, feature-krushank, feature-fiyans, feature-veer
Live Cloud Deployment	**https://dineflowai.duckdns.org** (AWS EC2 Ubuntu 24.04 LTS)

2. Team Member Code Contribution & Task Allocation Breakdown

Member Name	Git Branch	Engineering Module	Commits	Share
PATEL SHUBH M. Lead - 246310307123	feature-teamlead & main	Architecture, Laravel 11 Backend API, Cloud POS, Auth & Cloud Deploy	**612**	**40%**
DAVE KRUSHANK J. Member - 246310307028	feature-krushank	Cashier Web POS Screen, F1-F9 Shortcuts, Hardware Interfacing	**385**	**25%**
PATEL FIYANS B. Member - 246310307090	feature-fiyans	Admin Inventory ERP, Recipe BOM Depletion Logic, QR Generator	**305**	**20%**
PRAJAPATI VEER H. Member - 246310307147	feature-veer	Customer Web Cart Drawer, Mobile UI Layouts, KDS Stepper, QA	**230**	**15%**

3. Key Chronological Engineering Milestone Commits

Commit	Date	Author	Engineering Deliverable Summary
bc8c25e	2026-10-24	**Shubh Patel**	feat(architecture): complete multi-tenant cloud POS, licensing & security suite
770304e	2026-10-20	**Shubh Patel**	refactor(branding): unify DineFlow AI branding across views, controllers & DB
5fe6564	2026-08-19	**Krushank Dave**	feat(menu): implement customer QR interactive menu and admin menu manager
2f4534c	2026-08-16	**Fiyans Patel**	feat(desktop-db): implement embedded SQLite database manager & table repositories
84295a3	2026-08-16	**Veer Prajapati**	feat(sync): implement real-time offline sync engine, order queue & network monitor
ea5878e	2026-07-28	**Krushank Dave**	feat(frontend): add initial customer food category browsing and responsive UI

Project Guide Verification & Git Contribution Approval:

Prof. Yagnesh Patel

Lecturer, Department of Computer Engineering, K. D. Polytechnic, Patan

I have reviewed the GitHub commit history, branch distribution, and module allocations. The authentic engineering contributions of all 4 team members are hereby verified.

Guide Signature:

Date: _____ / _____ / 2026